

Harvard Extension Data Science

Dynamic Modeling and Forecasting in Big Data

Instructor: William Yu

Final Project

- **Project Goal:** The objective of this final project is to apply time series and/or dynamic models to a real-world dataset and generate meaningful predictions.
- **Topic Selection:** You are responsible for choosing your project topic. Select a topic based on your personal interests, career goals, or relevant work experience.
- **Data Requirement:** You may use any dataset, and larger datasets are encouraged
 - The only requirement is that the dataset must include a time-related variable—such as year, month, day, or intra-day frequency.
 - If you're having difficulty finding data, consider the following sources. Additional links are provided on the next page:
 - <https://www.kaggle.com/datasets>
 - <https://github.com/awesomedata/awesome-public-datasets>
- You *cannot* submit a project that has already been completed or submitted for another class.
- A group project is allowed. All group members will receive the same grade.

Three Components of the Final Project

(1) PowerPoint Presentation (Optional for Undergraduate Credit)

Prepare a slide deck of no more than 10 slides and deliver a 10-minute presentation. You may choose one of the following presentation formats:

- **Live presentation** on the final instruction day. Sign-up is required, as only 10 slots are available. A participation bonus will be awarded for live presentations.
- **Pre-recorded presentation** using one of the following methods:
 1. Use PowerPoint's recording function and submit the recorded slides. Do not submit a video file, as the file size may be too large.
 2. Upload your video recording to YouTube and show your YouTube weblink in the report.

(2) Written Report

Your written report should include the following elements:

- **Project purpose and motivation:** Explain the objective and significance of your analysis.
- **Visualizations:** Present clear, high-quality graphs and charts.

- **Model selection and justification:** Describe the models you used and explain why you chose them. Include test-set or out-of-sample validation.
- **Narrative and insights:** Tell a compelling story and summarize your key findings and takeaways.

Length: 5–40 pages, including charts and figures.

(3) Code

Submit your code in the form of an R script, R Markdown file, Python code, or Jupyter Notebook. If you prefer to combine the written report and code into a single R Markdown or Jupyter Notebook, that is acceptable. In that case, please submit it as an HTML file.

Submission Details: Submit all project materials, including slides, report, code, and dataset, via Canvas by midnight on **May 14**.

Your submission will be shared with the class on Canvas for educational purposes. If your data or report is proprietary and cannot be shared, please note this clearly in your report, and I will not share it with the class. If your dataset is large, for example larger than 50MB, you do not need to submit the data file itself; instead, provide the data source or website link.

Grading: Grading will be based primarily on the components outlined above in the written report and presentation. Strong projects typically feature:

1. A compelling and well-structured story
2. An interesting question and/or complicated, big, or valuable data
3. Robust and thoughtful data analysis
4. A novel or creative approach in topic, methodology, or application

Useful Links and Public Datasets

Use official data portals whenever possible. Below are suggested sources that are especially useful for time series, forecasting, panel data, text data, and large-scale public data projects.

General open-data and project idea sources

Kaggle Datasets - <https://www.kaggle.com/datasets>

Awesome Public Datasets (GitHub) - <https://github.com/awesomedata/awesome-public-datasets>

Data.gov (U.S. open data portal) - <https://www.data.gov/>

Our World in Data - <https://ourworldindata.org/>

UCI Machine Learning Repository - <https://archive.ics.uci.edu/>

Macroeconomic, finance, and business data

FRED (Federal Reserve Economic Data) - <https://fred.stlouisfed.org/>

Bureau of Economic Analysis (BEA) - <https://www.bea.gov/>

Bureau of Labor Statistics (BLS) - <https://www.bls.gov/>
U.S. Census Bureau data and APIs - <https://www.census.gov/data.html> and <https://www.census.gov/data/developers/data-sets.html>
World Bank Open Data / DataBank - <https://data.worldbank.org/> and <https://databank.worldbank.org/>
IMF Data portal - <https://data.imf.org/>
OECD Data Explorer - <https://data-explorer.oecd.org/>
Yahoo Finance historical market data - <https://finance.yahoo.com/>
SEC EDGAR filings - <https://www.sec.gov/edgar/search-and-access>
NBER data - <https://www.nber.org/research/data>

Housing, real estate, and prices

Federal Housing Finance Agency (FHFA) House Price Index - <https://www.fhfa.gov/data/hpi>
Zillow Research Data - <https://www.zillow.com/research/data/>
S&P Dow Jones Indices / Case-Shiller resources - <https://www.spglobal.com/spdji/en/>
BIS Property Price Statistics - <https://www.bis.org/statistics/pp.htm>
Dallas Fed International House Price Database - <https://www.dallasfed.org/institute/houseprice>
Robert Shiller data - <https://shillerdata.com/>

Climate, environment, and energy

NOAA Climate Data Online - <https://www.ncei.noaa.gov/cdo-web/>
NASA POWER climate and solar data - <https://power.larc.nasa.gov/>
U.S. Energy Information Administration (EIA) - <https://www.eia.gov/opensdata/>

Large-scale text, search, and alternative data

Google Trends - <https://trends.google.com/>
GDELT Project (global news and events data) - <https://www.gdeltproject.org/>
Google Books Ngram Viewer - <https://books.google.com/ngrams/>
OpenAlex (scholarly metadata) - <https://openalex.org/>

International and long-run historical data

Penn World Table - <https://www.rug.nl/ggdc/productivity/pwt/>
Maddison Project Database - <https://www.rug.nl/ggdc/historicaldevelopment/maddison/>
Gapminder - <https://www.gapminder.org/data/>
CIA World Factbook - <https://www.cia.gov/the-world-factbook/>

Data access tools students may find helpful

Cross Validated (statistics Q&A) - <https://stats.stackexchange.com/>
Stack Overflow - <https://stackoverflow.com/>
Harvard Dataverse - <https://dataverse.harvard.edu/>
Google Dataset Search - <https://datasetsearch.research.google.com/>